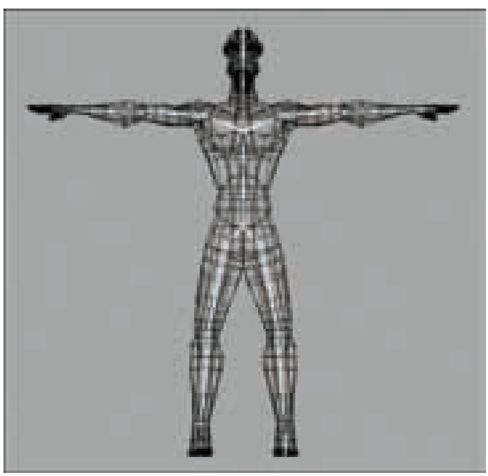
points move when the rig moves. The muscles are special deformer lattices being moved by the skeleton and in turn deforming the model to make more realistic movement. Skeletons consist of articulated joints and bones.

- **Adding joints and bones:** Joints are places where your model can be bent or rotated. You can create a joint using Joint tool in any popular 3D modeling software. If you create a joint and then create another joint, a bone and a joint will be created as a child of first joint. This auto-parenting is an important feature of rigs and is crucial. Bones are just connections between joints. Rigging (Creating a skeletal rig): To make a realistic skeletal rig, it is important to understand the anatomy of the character/object you are modeling and then placing joints at appropriate places. For example if you put shoulder joint too close to the neck, the arm will disappear into the body when rotating downwards. If you put it too far away, it will show a gap between the arm and body. Once joints and bones are in place, you decide the bind pose. This pose is a starting position that you can easily go back to, with each joint rotation and transformation set to zero.
- **Skinning:** Once you are done with skeletal rigging and aligning it with your model, you need to connect the rig to your model to move the model surface. This is called binding or skinning. The simplest way to have a rig move a surface is to parent the surface to the rig. There are other methods too such as rigid binding and smooth binding.



Rigging

Tools used

- 3DS Max and 3DS Max Design
- Blender(Free)
- Lightwave : Student Price: $195
- Maya
- Softimage